The complete quadrilateral in Lean 4: Newton–Gauss, Bodenmiller, the Miquel point, the Steiner circle and the Simson line

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Abstract

Four lines in general position form a complete quadrilateral: six vertices, three diagonals, and four triangles, one for each line omitted. We give machine-checked proofs, in Lean 4 with Mathlib, of eight classical theorems about this configuration, all from the single hypothesis that the lines are in general position. On the side of the diagonals: the midpoints of the three diagonals are collinear (the Newton–Gauss line); the orthocentres of the four triangles are collinear; every orthocentre has equal power with respect to the three circles on the diagonals as diameters (the Gauss–Bodenmiller theorem); and the line of the orthocentres is perpendicular to the Newton–Gauss line. On the side of the circles: the circumcircles of the four triangles pass through a common point M, given by an explicit formula (the Miquel point); the four circumcentres and M lie on one circle (the Steiner, or Miquel, circle); the feet of the perpendiculars from M to the four lines are collinear (the Simson line of the quadrilateral); and that line is parallel to the line of the four orthocentres, as Mathlib affine subspaces (whose parallelism allows the two lines to coincide). The two halves are separate developments that share one file of definitions. We also show that two of the four orthocentres can coincide in general position (two of them coincide exactly when two of the lines are perpendicular), so the orthocentre line is determined only because the four are never all equal, which we prove. The local definitions (line, intersection, circumcentre, orthocentre, foot of a perpendicular, power of a point, general position) are proved equal to Mathlib's. All proofs use only Lean's three standard axioms. Version 2 adds two checks: every main theorem is restated with Mathlib-only types and shown non-vacuous on two explicit configurations, and all twelve Lean statements written by another model from textbook statements alone were proved from the developments, two of them with new proofs. In the sources we searched, the Newton–Gauss line and the four-line Miquel theorem already have formal proofs (in HOL Light and Rocq); we found none of the other six in any system, and none of the eight in Lean.

1 Introduction

A *complete quadrilateral* is a set of four lines in the plane, no two parallel and no three through a point. Its six intersection points are its *vertices*; the three segments joining the pairs of vertices that

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do not lie on a common line are its *diagonals*; and omitting one line at a time leaves four triangles. The configuration carries a remarkable family of classical theorems, found between Gauss and Steiner in the first third of the nineteenth century [1, 2, 3]. Two of them concern points: the midpoints of the diagonals lie on a line, and so do the four orthocentres. Two concern circles: the circles on the diagonals as diameters are coaxal, and the four circumcircles pass through one point. The rest tie the two families together. We formalise eight of them (Table 1).

	theorem	Lean	source (pinned)
1	midpoints of the diagonals collinear	`QuadNewton.newton_gauss`	`Quad`:53
2	four orthocentres collinear (not all equal)	`QuadNewton.orthocentres_collinear`	`Quad`:79
3	diagonal circles coaxal (equal power)	`QuadNewton.bodenmiller`	`Quad`:135
4	orthocentre line $\perp$ Newton–Gauss line	`QuadNewton.orthocentre_line_perp`	`Quad`:155
5	four circumcircles concur at M	`QuadMiquel.miquel`	`Miquel`:162
6	circumcentres and M concyclic	`QuadMiquel.steiner_circle`	`Miquel`:200
7	feet from M to the four lines collinear	`QuadMiquel.simson`	`Simson`:155
8	that line $\parallel$ the orthocentre line	`QuadMiquel.steiner_parallel_line`	`Parallel`:241

Table 1: The eight theorems. Each assumes only that the four lines are in general position.

1.1 Contributions

- One model of the complete quadrilateral in $\mathbb{C}$, with explicit formulas for the intersection point of two lines, the circumcentre, the orthocentre and the foot of a perpendicular (Section 2), shared by two independent developments.
- The eight theorems, each stated for arbitrary four lines with the single hypothesis `GenPos` (Sections 3 and 4). The collinearity and concyclicity conclusions use Mathlib's `Collinear` and `EuclideanGeometry.Cospherical`.
- An explicit formula for the Miquel point, as the centre of a spiral similarity (Section 4).
- A small finding about the statements themselves: in general position two of the four orthocentres can coincide; $H_3 = H_4$ exactly when $L_1 \perp L_2$ (`QuadMiquel.ortho34_eq_iff`). So "the line of the orthocentres" needs the fact, which we prove, that they are not all equal (Section 3), and Steiner's parallelism, stated for a fixed pair of orthocentres, needs that pair to be distinct (Section 4).
- Bridges tying every local notion to Mathlib's (Section 5), and the uniqueness of the Miquel point: any point on the circumcircles of T_4, T_3 and T_1 is M (`QuadMiquel.miquelPt_unique`); in particular the common point of all four is unique.
- In version 2 (Section 7.1): Mathlib-only restatements of all ten Lean statements of the eight theorems, each with two explicit witnesses, and proofs from the developments of twelve Lean statements written by another model from textbook statements.

What is not claimed. We prove each theorem for one fixed labelling of the four lines; the statements are symmetric in content but we did not prove the symmetry. The diagonal circles are shown to be coaxal in the sense that the four orthocentres have equal power with respect to all three, which together with Theorem 2 places the radical axis; we do not formalise coaxal pencils as such. (That the set of points of equal power with respect to two of the circles is exactly a line

is not stated in the main files; it is proved in the Layer-15 file, Section 7.1.) Steiner's parallelism is proved in Mathlib's sense, which allows the Simson line and the orthocentre line to coincide; that they are distinct is not proved.

1.2 Availability

The proofs are in two Lean 4 developments on Lean v4.33.1 with Mathlib pinned at revision `0df444a3`. File and line counts are of the library `.lean` files, including the root import file and counting every physical line; in total 14 files and 2,980 lines. Supporting material (the verification battery reports, fresh-clone receipts, numerical scripts, blind re-formalisation files, reviews and hand-check sheet of version 1, and the Layer-14 and Layer-15 reports, the blind writer's statements, the transcript audit and the reviews of version 2) is in the folder `PaperE-complete-quadrilateral` of the repository `https://github.com/jpbald93/LEAN-verification`.

repository	content	files	lines
`quadnewton`	Theorems 1–4 and bridges	6	1,066
`quadmiquel`	Theorems 5–8 and bridges	8	1,914

The repositories are `https://github.com/jpbald93/quadnewton` at commit `48e94cf944b3dfbcca7358360a064e39b6336d98` and `https://github.com/jpbald93/quadmiquel` at commit `011c9ecd5276b7abaf2b42a8f5ba463196df123d`; Lean v4.33.1, Mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474` (pinned in `lake-manifest.json`). Source links in this paper point to these commits. Version 2 of this paper adds, in a new commit of each repository (`quadnewton` at commit `c3a4ce513eb8789f5f0de4ae02efe707a718466c`, `quadmiquel` at commit `4089a974d5f56663be3daa5203ec856c28b793c3`), the Layer-14 and Layer-15 files `QuadNewton/L14Newton.lean`, `QuadNewton/L15Classical.lean`, `QuadMiquel/L14Miquel.lean` and `QuadMiquel/L15Classical.lean` (Section 7.1), two import lines in each root file, and the extended gate lists and README. Apart from the two root import files, all earlier library files are unchanged (same SHA-256 hashes), and all earlier theorem statements are unchanged, so the source links above remain valid. With the new files the repositories have 8 files and 2,328 lines (`quadnewton`) and 10 files and 3,169 lines (`quadmiquel`); the table above counts the first version. Each repository has a gate script (`gate.sh`) that rejects forbidden tokens, builds the library and checks with `#print axioms` that every listed declaration depends only on `propext`, `Classical.choice` and `Quot.sound`, and a tampering suite (`tests/tamper.sh`) that plants defects the gate must reject.

2 The model

Both developments start from the same file `Basic.lean`, identical up to the name of its namespace. Points are complex numbers. For $a, b \in \mathbb{C}$ let $\mathrm{cr}(a,b) = \mathrm{Im}(\bar{a}b)$, the cross product of a and b as plane vectors. A line L is a pair (p,d) of a point and a direction, with carrier $\{z : \mathrm{cr}(d, z-p) = 0\}$. For lines $L = (p,d)$ and $M = (q,e)$,

$$\mathrm{meet}(L,M) = p + \frac{\mathrm{cr}(e, q-p)}{\mathrm{cr}(e,d)}\, d,$$

which is their intersection point when $\mathrm{cr}(e,d) \neq 0$. Four lines $L_1, \ldots, L_4$ are in *general position* (`GenPos`) if their directions are nonzero, no two are parallel ($\mathrm{cr}(d_i, d_j) \neq 0$), and no three pass through one point ($\mathrm{meet}(L_i, L_j) \notin L_k$ for the four triples). We write $A_{ij} = \mathrm{meet}(L_i, L_j)$ for the six

vertices; the diagonals are $A_{12}A_{34}$, $A_{13}A_{24}$ and $A_{14}A_{23}$; and T_k is the triangle of the three vertices not on L_k: $T_4 = (A_{12}, A_{13}, A_{23})$, $T_3 = (A_{12}, A_{14}, A_{24})$, $T_2 = (A_{13}, A_{14}, A_{34})$, $T_1 = (A_{23}, A_{24}, A_{34})$.

The circumcentre, the orthocentre and the foot of a perpendicular are given by explicit formulas:

$$\mathrm{circ}(a,b,c) = a + \frac{|b-a|^2(c-a) - |c-a|^2(b-a)}{\overline{(b-a)}(c-a) - (b-a)\overline{(c-a)}}, \qquad \mathrm{ortho}(a,b,c) = a+b+c-2\,\mathrm{circ}(a,b,c),$$

$$\mathrm{foot}(z,L) = p + \frac{\mathrm{Re}\,(\bar{d}(z-p))}{|d|^2}\,d.$$

The orthocentre is defined through Euler's relation $H = A + B + C - 2O$. Section 5 proves that each of these is the corresponding Mathlib notion.

Normal coordinates. Both developments first prove that general position puts the configuration into a normal form. In `quadnewton` (`QuadNewton.normal_coords`): there are $O, u, w \in \mathbb{C}$ and real s, t with $A_{12} = O$, $A_{13} = O + u$, $A_{23} = O + w$, $A_{14} = O + su$, $A_{24} = O + tw$ and $A_{34} = O + \big(s(1-t)u + t(s-1)w\big)/(s-t)$, where $\mathrm{cr}(u,w) \neq 0$, $s, t \notin \{0,1\}$ and $s \neq t$. `quadmiquel` proves the same normal form independently, in its own notation. The central coordinate identities are then rational-function identities in $u, w, \bar{u}, \bar{w}, s, t$ with those non-vanishing conditions, discharged principally by `field_simp`, `ring` and `linear_combination` with the conjugates treated as independent variables; incidence and non-degeneracy lemmas connect them to the geometric conclusions. The two developments were written separately, by different agents on different machines, from the shared definitions.

3 The diagonals and the orthocentres

All four theorems of this section are in `quadnewton`.

Theorem 1 (Newton–Gauss line; `QuadNewton.newton_gauss`)**.** *If $L_1, \dots, L_4$ are in general position, the midpoints of the three diagonals are collinear; and the midpoints of $A_{12}A_{34}$ and $A_{13}A_{24}$ are distinct (`QuadNewton.diag_midpoints_ne`).*

Theorem 2 (Orthocentre line; `QuadNewton.orthocentres_collinear`, `QuadNewton.orthocentres_not_all_eq`)**.** *If $L_1, \dots, L_4$ are in general position, the orthocentres $H_1, \dots, H_4$ of $T_1, \dots, T_4$ are collinear, and they are not all equal: $H_3 \neq H_4$ or $H_2 \neq H_4$.*

The second part is needed, and cannot be strengthened to a fixed pair. For L_1 the real axis, L_2 the imaginary axis, L_3 the line through 1 and i, and L_4 the line through 2 and $-2i$, the lines are in general position and $H_1 = H_2 = \frac{3}{2} - \frac{1}{2}i$, $H_3 = H_4 = 0$ (checked in exact rational arithmetic). So any one pair of orthocentres can coincide, and "the line of the orthocentres" is determined only because the four are never all equal.

For points $P \neq Q$ let $\mathrm{pw}(z; P, Q) = |z - \frac{P+Q}{2}|^2 - \frac{1}{4}|P-Q|^2$, the power of z with respect to the circle on PQ as diameter (it is Mathlib's `EuclideanGeometry.Sphere.power` of that circle: `QuadNewton.pw_eq_power`).

Theorem 3 (Gauss–Bodenmiller; `QuadNewton.bodenmiller`)**.** *If $L_1, \dots, L_4$ are in general position, each orthocentre H_k has the same power with respect to the three circles having the diagonals as diameters.*

Since the centres of these circles (the diagonal midpoints) are pairwise distinct (`QuadNewton.bodenmiller_coaxal`, which also restates Theorem 3 with Mathlib's power), the points of equal power with respect to two of them form a line (an elementary fact not stated in the main files; it is proved in the Layer-15 file as `QuadNewton.Classical.radical_eq_line`, Section 7.1), and Theorem 3 says all four orthocentres lie on the radical axis of every pair; that is the classical statement that the three circles are coaxal, with the orthocentre line as their radical axis.

Theorem 4 (`QuadNewton.orthocentre_line_perp`)**.** *If $L_1, \dots, L_4$ are in general position, then for any two orthocentres H, H', $\operatorname{Re}\left((m_2 - m_1)\overline{(H' - H)}\right) = 0$, where m_1, m_2 are the midpoints of $A_{12}A_{34}$ and $A_{13}A_{24}$.*

With $m_1 \neq m_2$ (Theorem 1) and two distinct orthocentres (Theorem 2), this is the perpendicularity of the orthocentre line and the Newton–Gauss line. When $H = H'$ the statement holds trivially. The real part is Mathlib's inner product on $\mathbb{C}$ (`QuadNewton.perp_iff_inner`; the inner-product form is `QuadNewton.orthocentre_line_perp_inner`).

4 The circles

All four theorems of this section are in `quadmiquel`. The Miquel point is given explicitly:

$$M = A_{12} + \frac{(A_{13} - A_{12})(A_{24} - A_{12}) - (A_{23} - A_{12})(A_{14} - A_{12})}{A_{13} + A_{24} - A_{23} - A_{14}}$$

(`QuadMiquel.miquelPt`), the centre of the spiral similarity taking $A_{13} \mapsto A_{14}$ and $A_{23} \mapsto A_{24}$. General position makes the denominator nonzero. M is the reflection of A_{12} in the line through the circumcentres of T_4 and T_3, so it is the second intersection point of those two circles; it is none of the six vertices; and it is the only point on the circumcircles of T_4, T_3 and T_1 (`QuadMiquel.miquelPt_spec`, `QuadMiquel.miquelPt_ne_vertices`, `QuadMiquel.miquelPt_unique`).

Theorem 5 (Miquel point; `QuadMiquel.miquel`)**.** *If $L_1, \dots, L_4$ are in general position, M lies on the circumcircle of each of $T_1, \dots, T_4$ (the circle with centre $\operatorname{circ}(T_k)$ through the vertices of T_k).*

Theorem 6 (Steiner circle; `QuadMiquel.steiner_circle`)**.** *If $L_1, \dots, L_4$ are in general position, the four circumcentres $\operatorname{circ}(T_1), \dots, \operatorname{circ}(T_4)$ and M are cospherical (lie on one circle).*

Theorem 7 (Simson line of the quadrilateral; `QuadMiquel.simson`)**.** *If $L_1, \dots, L_4$ are in general position, the feet of the perpendiculars from M to $L_1, \dots, L_4$ are collinear.*

Classically this follows from the triangle Simson line: M is on the circumcircle of each T_k, so the three feet on the sides of T_k are collinear, and the four triples overlap. Our proof is direct. The four feet are pairwise distinct (`QuadMiquel.simson_feet_ne`), so the line is determined.

Theorem 8 (Steiner; `QuadMiquel.steiner_parallel_line`)**.** *If $L_1, \dots, L_4$ are in general position, with F_k the foot of the perpendicular from M to L_k, the line through F_1 and F_2 is parallel (`AffineSubspace.Parallel`) to the affine span of the four orthocentres $H_1, \dots, H_4$.*

The line through F_1, F_2 is the Simson line (Theorem 7; $F_1 \neq F_2$), and the affine span of the orthocentres is the orthocentre line: within this development we also prove that the four are collinear and not all equal (`QuadMiquel.orthocentres_collinear'`, `QuadMiquel.orthocentres_not_all_eq'`). Mathlib's `AffineSubspace.Parallel` includes the case of equal lines; we do not prove that the Simson line differs from the orthocentre line. The proof goes through the cross-product relations $\operatorname{cr}(F_2 - F_1,\ H_i - H_4) = 0$ for $i = 1, 2, 3$ (`QuadMiquel.steiner_parallel_all`).

A fixed pair of orthocentres would not do. $H_3 = H_4$ exactly when $L_1 \perp L_2$ (`QuadMiquel.ortho34_eq_iff`): then both T_3 and T_4 have their right angle at A_{12}, and both orthocentres are A_{12}; the example after Theorem 2 is of this kind. The first version of the theorem we proved, `QuadMiquel.steiner_parallel_affine`, used the line through H_4 and H_3 and needed the extra hypothesis $L_1 \not\perp L_2$. The theorem above needs none, because when $H_3 = H_4$ both H_1 and H_2 differ from it (`QuadMiquel.ortho12_ne_of_eq34`); $H_1 = H_2$ can still happen.

5 Bridges to Mathlib

Each local notion of Section 2 is tied to the standard one, in both developments (file `Bridge.lean`; the `quadnewton` names are given, and `quadmiquel` has equivalent statements under the same names, with two packaging differences: its `genPos_triangles` gives only the non-collinearity, the vertex distinctness being the separate `QuadMiquel.genPos_vertices_ne`, and its `genPos_iff` states non-concurrence with the local carriers, which `carrier_eq_line` identifies with Mathlib's lines):

- a line with nonzero direction is Mathlib's line through p and $p+d$ (`QuadNewton.carrier_eq_line`), and meet(L, M) is the unique common point of two non-parallel lines (`QuadNewton.meet_eq_iff`);
- for non-collinear a, b, c, circ and ortho are Mathlib's `Affine.Simplex.circumcenter` and `Affine.Triangle.orthocenter` of the triangle (`QuadNewton.circ_eq_circumcenter`, `QuadNewton.ortho_eq_orthocenter`);
- for a nonzero direction, foot(z, L) is Mathlib's orthogonal projection of z onto the line (`QuadNewton.foot_eq_orthogonalProjection`);
- general position makes each of the four triangles non-degenerate and the six vertices pairwise distinct (`QuadNewton.genPos_triangles`), so the previous bridges apply to every triangle the theorems use; and `GenPos` is equivalent to its intended meaning stated with Mathlib's notions (nonzero directions, no two of the lines `AffineSubspace.Parallel`, no three with a common point: `QuadNewton.genPos_iff`).

Each bridge was first checked numerically, including on forced degenerate inputs; the only statement that failed as first written was the fixed-pair parallel form `QuadMiquel.steiner_parallel_affine` (Section 4), which led to Theorem 8.

6 Related work

Classical. The theorems are classical; standard accounts are Johnson [3] (where the coaxality of the diagonal circles appears as the Gauss–Bodenmiller theorem, p. 172) and Coolidge [2]. Several of them, including the Miquel point and circle of the complete quadrilateral, go back to Steiner's 1828 note on the complete quadrilateral [1]. We do not attempt finer historical attributions.

Formal proofs. On 14 October 2026 we searched for each of the eight theorems in Mathlib (code, `docs/100.yaml` and `docs/1000.yaml`, issues and pull requests, both at the pinned revision and at the current master), the Archive of Formal Proofs (a full-text search of a mirror of 13 October 2026), all 1500 articles of the Mizar library, HOL4, HOL Light, Metamath's `set.mm`, Rocq's GeoCoq, HighSchoolGeometry, the area-method library and the projective-geometry library, ProofAtlas (partly: sitemap and search index), the public archive of the Lean Zulip, compfiles,

`formal-conjectures`, FormalBook, the lean-genius repository, and GitHub code search. Two of the eight have complete formal proofs:

- the Newton–Gauss line, as `GAUSS` in HOL Light's `Complex/grobner_examples.ml` (Chou's example 15 [4], proved by a Gröbner-basis procedure) [5], and as `gauss_line` in the Rocq area-method library [6];
- the four-line Miquel theorem, as `Miquel` in Rocq's HighSchoolGeometry [7, 8], in conditional form: under the library's incidence and non-collinearity hypotheses (which exclude, for instance, the shared vertex itself), a point on two specified circumcircles lies on the other two. It does not construct the common point; our Theorem 5 gives it by an explicit formula.

We found no formal proof, in any system, of Theorems 2, 3, 4, 6, 7 or 8, and none of the eight in Lean. The triangle versions of nearby theorems are formalised: the triangle Simson line (AFP, HOL Light, HighSchoolGeometry, lean-genius), the triangle Miquel (pivot) theorem (AFP [9], lean-genius), and the triangle Steiner line (AFP). A synthetic engine (Newclid/JGEX) also proves the four-line Miquel theorem, but its proofs are not checked by a proof assistant. The search was run by one agent; we re-read the three hits (`GAUSS`, `gauss_line` and `Miquel`) at their sources and confirmed that they state the theorems described. These are completed proofs relative to their libraries' own foundations (the area-method library works over an axiomatised field with the Chou–Gao–Zhang area axioms, and HighSchoolGeometry over an axiomatic plane); we inspected them but did not rebuild them, and the area-method library contains three `Admitted` lemmas which the sweep judged to be outside the path of `gauss_line`, a judgement we could not confirm with `Print Assumptions`.

This series. This is the fifth paper in a series of Lean formalisations of classical plane geometry by coordinate certificates [10, 11, 12, 13]. Paper A proves the triangle Simson line; Paper D Clifford's chain for four to six circles through a common point. Classically, inversion at that common point turns the four-circle case into the four-line Miquel theorem; we did not formalise this correspondence, and the two developments are independent.

7 Verification

Gates and tampering. After publishing, both repositories were cloned fresh from GitHub at the pinned commits on both machines (16 October 2026, UTC): an x86-64 Linux machine and an arm64 macOS machine, with Lean v4.33.1 and the Mathlib revision above. On each, the checkout was clean, the build succeeded, the gate passed (`quadnewton`: 35 declarations; `quadmiquel`: 62; standard axioms only) and every planted defect was rejected; the SHA-256 hashes of all 14 library files agree between the two machines and with the published trees. The receipt logs and script are in the verification repository. Before publishing, each development was gated and tamper-tested on both machines: an x86-64 Linux machine and an arm64 macOS machine. `quadnewton`: 35 declarations, `quadmiquel`: 62 declarations, all with standard axioms only; every planted defect rejected (six tests each: a hidden `sorry`, an axiom split across two lines, a fake `#print axioms` report, a command-redefining macro, a wrapper axiom, and the removal of one needed hypothesis).

Numerical re-check. A script (in the verification repository) evaluates each conclusion at 60 significant digits on 1000 random configurations (seed 20261014), each with all pairwise direction cross products and all vertex-to-other-line distances at least 10^{-2}:

theorem	worst residual
1 Newton–Gauss	4.7×10^{-60}
2 orthocentres collinear	2.4×10^{-59}
3 Bodenmiller (power differences)	1.3×10^{-54}
4 perpendicularity (normalised)	8.9×10^{-60}
5 Miquel (relative)	1.5×10^{-57}
6 Steiner circle (relative)	6.3×10^{-58}
7 Simson line	1.2×10^{-57}
8 parallelism, all $H_i - H_4$ (normalised)	1.5×10^{-58}

A separate run of 300 configurations with $L_1 \perp L_2$ forced (same margins) checks the exceptional case: there $|H_3 - H_4| \leq 7.5 \times 10^{-59}$, both H_1 and H_2 stay at distance at least 0.16 from H_3, and the parallelism residual over all $H_i - H_4$ is at most 2.7×10^{-59}. Three negative controls failed in every trial, with smallest violations 2.7×10^{-7} (Newton–Gauss with one midpoint of the configuration and two of a perturbed copy, L_1 shifted by 0.1), 1.1×10^{-8} (the unperturbed M on the perturbed circumcircles) and 3.2×10^{-3} (feet from a point other than M). Numerical checks are evidence, not proofs.

Statement-level battery. Both developments, with their bridge files, were run through the battery used in the earlier papers, on both machines (15 October 2026). The two machines agree in every category except satisfiability, where some declarations are certified on one machine and not the other.

- *Conclusion mutations.* `quadnewton` 149/149 and `quadmiquel` 293/294 rejected (two by the heartbeat limit). The one survivor, on both machines, flips the middle equation inside `QuadMiquel.orthocentres_not_all_eq'`, $\neg(H_1 = H_4 \wedge H_2 = H_4 \wedge H_3 = H_4)$. It survives because the proof never uses that conjunct: it shows the stronger $\neg(H_1 = H_4 \wedge H_3 = H_4)$, from `QuadMiquel.ortho12_ne_of_eq34`. The survivor marks a statement weaker than its proof, not an error. The generator alters only conclusions of certain forms (equalities, numeric literals), so several main theorems produced no mutants: `QuadNewton.orthocentres_collinear`, `QuadMiquel.miquel`, `QuadMiquel.steiner_circle`, `QuadMiquel.simson` and `QuadMiquel.steiner_parallel_line`.

- *Contradiction probe.* No contradiction derived from the hypotheses of any declaration (40 and 108 probed). This does not show that the hypotheses are consistent.

- *Satisfiability.* Certified for 48 (x86-64) or 51 (arm64) of 69 declarations of `quadnewton` and 76 or 89 of 139 of `quadmiquel`; every main theorem is among the unresolved. The numerical re-check exhibits configurations in general position for each.

- *Hypothesis ablation.* Each hypothesis binder is removed in turn and the unchanged proof re-checked; a failure shows that this proof uses the binder, not that it is logically necessary. In `quadnewton` removing any of the 51 tested binders broke the proof. In `quadmiquel` 204 broke it and 17 did not: all 17 are non-vanishing conditions in auxiliary algebraic lemmas (`circC_T3`, `circC_T4`, `circC_scaled`, `simF2`, `simF3`, `simF4`). Some are superfluous because the identity extends to the zero-denominator case under Lean's total division, the others because the term they protect cancels from the identity. No hypothesis of a main theorem is unused. We keep them, since they are harmless and removing them would change the line numbers this paper cites.

- *Linter.* No unused-variable warnings in the library files.

Does the statement mean what it says? Layers 1–8 do not show that a formal statement means what the prose says. To test that, we gave a different language model from the one that wrote the proofs only the text of this paper with every code name, source link and the verification material removed. From the prose alone it wrote ten Lean statements, one for each part of Theorems 1–8, with its own definitions taken from the formulas of Section 2. Two separate comparer runs, using the same model as the proof author (the blind writer was a different model), then proved in Lean, for each of the ten, that the blind statement implies ours and ours implies it, using only the premise, unfolding of definitions, and general Mathlib lemmas (no theorem of our developments in any proof body). All twenty implications were proved, with standard axioms only; no difference was found. The comparisons needed only definitional agreements between the two sets of definitions, plus one rewriting of the power of a point (normSq versus the squared norm). This check is only as independent as the blind text allows: the model section gives the defining formulas, so it tests the statements of the theorems, not the choice of definitions, which Section 5 addresses. The writer's claim to have read only the blind text was checked by the author against its full transcript; the complete arguments of all six tool calls it made (two reads of the blind text; shell commands that write and compile its own file) are in the verification repository, with machine names redacted.

Statements checked by hand. The author read each of the eight theorems as given in this paper against its ten Lean statements at the published commits (a sheet pairing the two, with links to the source lines, is in the verification repository) and found them to agree. This is a reading check, not a proof.

7.1 Layers 14 and 15

These checks were added in version 2. The results of the earlier paragraphs refer to the first-version files.

Layer 14: statements in Mathlib's language and non-vacuity. A Layer-14 file in each repository (`QuadNewton/L14Newton.lean`, `QuadMiquel/L14Miquel.lean`) restates each of the ten Lean statements of Theorems 1–8 with Mathlib-only types and proves it from the library theorem through bridges (`QuadNewton.l14_newton_gauss`, `QuadNewton.l14_diag_midpoints_ne`, `QuadNewton.l14_orthocentres_collinear`, `QuadNewton.l14_orthocentres_not_all_eq`, `QuadNewton.l14_bodenmiller`, `QuadNewton.l14_orthocentre_line_perp`, `QuadMiquel.l14_miquel`, `QuadMiquel.l14_steiner_circle`, `QuadMiquel.l14_simson`, `QuadMiquel.l14_steiner_parallel_line`); none was left out. Lines are `line[ℝ,p,q]`, general position is stated with `AffineSubspace.Parallel` and the absence of a common point of three lines, vertices are arbitrary common points, the triangles are arbitrary `Affine.Triangle`s with the given vertices (`QuadNewton.l14_triangles_exist`, `QuadMiquel.l14_triangles_exist` show that they exist), the Miquel point is *any* point on the four Mathlib circumspheres (with an ∃! statement), the power is `EuclideanGeometry.Sphere.power` and the feet are `EuclideanGeometry.orthogonalProjection`. The distinctness statement of Theorem 1 is restated for all three pairs of midpoints (the library states one pair; the other two follow from `QuadNewton.bodenmiller_coaxal`). A mechanical check of the elaborated types of these statements, the two existence statements and the witnesses (32 declarations) found no constant of our namespaces. Each statement has two witness theorems that prove every hypothesis on an explicit configuration, one generic and one with $L_1 \perp L_2$ (20 witnesses), and prove the conclusion non-degenerate: the three midpoints pairwise distinct; the four orthocentres pairwise distinct (in the perpendicular case $H_3 = H_4$, with H_1, H_2, H_3 pairwise distinct); explicit common powers and

distinct circle centres; nonzero direction vectors for the perpendicularity; M different from the six vertices, with the four circumcentres pairwise distinct; the five points of the Steiner circle pairwise distinct; the four feet pairwise distinct; and for Steiner's parallel two distinct feet and a genuine orthocentre line, also when $H_3 = H_4$. No Mathlib-only statement turned out false or needed an extra hypothesis.

Layer 15: comparison with the classical statements. For this paper the classical statements E1–E8 were written from the standard literature by the orchestrating agent (not collected by a separate agent with page citations): the Newton–Gauss line with non-collapse, the orthocentre line with non-collapse, Gauss–Bodenmiller with the radical axis, the perpendicularity, Miquel, the Steiner circle, the Simson line and Steiner's parallel. A writer of a different model family (GPT-based), given only those statements and a list of Mathlib names, and no code or text of this paper, wrote them in Lean with Mathlib notions and its own definitions: 12 statements and 2 construction obligations (the six vertices exist uniquely; the triangles are affinely independent), all with `sorry`, 0 errors. It marked two ambiguities ($\exists$ rather than $\exists!$ for the Miquel point; parallelism allowing coincidence) and no extra hypothesis. By its own report it read, besides its brief and its own files, one generic procedure file with no content of this paper; this departs from the brief-only rule and is disclosed. A comparer (an Opus model) copied the definitions and statements verbatim (checked mechanically) and proved each from the developments, with no added hypothesis and no change to the library; both construction obligations were proved. Of the 12: 10 covered outright, 2 narrower in the library but fully proved by new proofs, none different, none out of scope, and no writer statement false as written.

- *New proof* E3(b), `QuadNewton.Classical.cov_E3_b`: one line contains the orthocentres and is the entire equal-power locus of every pair of diagonal circles. The library gives equal powers and distinct centres, not that the locus is exactly a line; new lemma `QuadNewton.Classical.radical_eq_line`.
- *New proof* E7(b), `QuadMiquel.Classical.cov_E7_b`: the four feet from *any* point are not all equal. The library proves distinct feet only from M (`QuadMiquel.simson_feet_ne`, giving `QuadMiquel.Classical.cov_E7_b_part`); a common foot would lie on three lines.
- *Ambiguities.* The stronger $\exists!$ reading of E5 is also proved (`QuadMiquel.Classical.cov_E5_a_unique`). For E8 only Mathlib's parallelism is proved (`QuadMiquel.Classical.cov_E8_a`); strict parallelism (distinct lines) is not.

The other covered statements are `QuadNewton.Classical.cov_E1_a`, `_E1_b`, `_E2_a`, `_E2_b`, `_E3_a`, `_E4_a` and `QuadMiquel.Classical.cov_E5_a`, `_E6_a`, `_E7_a`. They are collected in `QuadNewton/L15Classical.lean` (E1–E4, namespace `QuadNewton.Classical`) and `QuadMiquel/L15Classical.lean` (E5–E8, namespace `QuadMiquel.Classical`). With the Layer-14 and Layer-15 files the gates pass with 81 (`quadnewton`) and 104 (`quadmiquel`) declarations, standard axioms only; the tampering suites rejected every planted defect, and the earlier library files other than the root import files are unchanged.

8 Limitations

1. **One labelling.** Each theorem is proved for the given order of the lines; the symmetric versions are not stated.

2. **Local notions.** The main theorems are stated with the local definitions; the bridges of Section 5 make them statements about Mathlib's notions, and in version 2 the Layer-14 files restate every one of them with Mathlib-only types (Section 7.1).

3. **Coaxality** is stated through equal powers, not as a pencil of circles; that the radical locus of two diagonal circles is exactly a line is proved only in the Layer-15 file.

4. **Parallel, not strictly parallel.** Theorem 8 uses Mathlib's parallelism, which allows the two lines to coincide; their distinctness is not proved.

5. **Prior art.** Two of the eight theorems have earlier formal proofs in other systems.

Acknowledgements and use of AI

A large-language-model AI assistant was used to write the Lean proofs, to run the prior-art searches and the verification battery, and to draft this paper. The author directed the work and reviewed the paper. Every proof is checked by the Lean kernel, and the author takes responsibility for all content.

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